

AI Agent f



eraction

Analysis

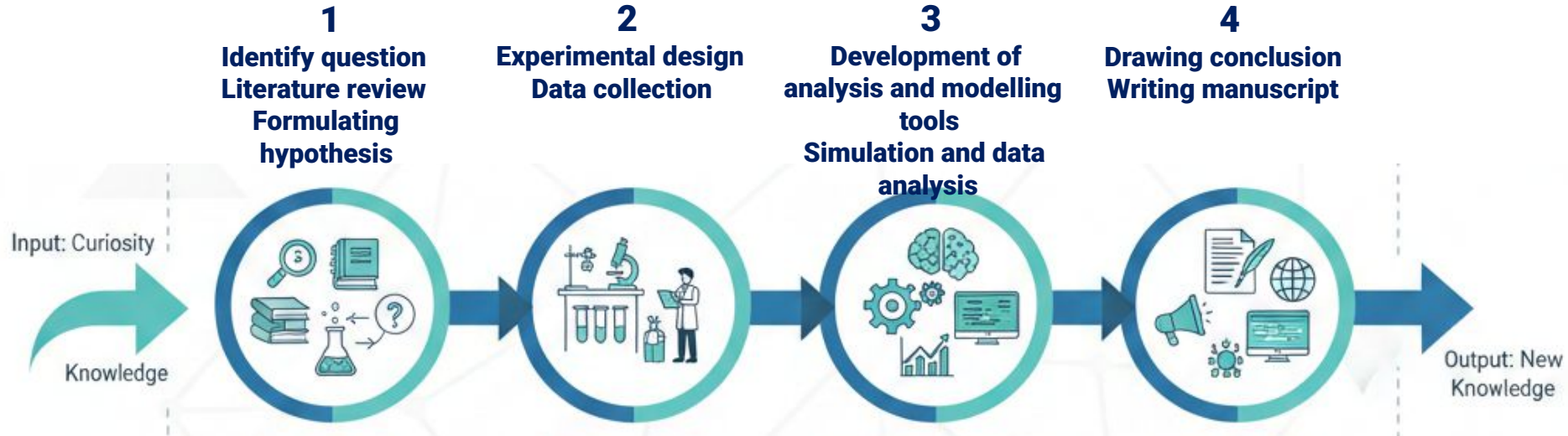


Andrea Brovelli

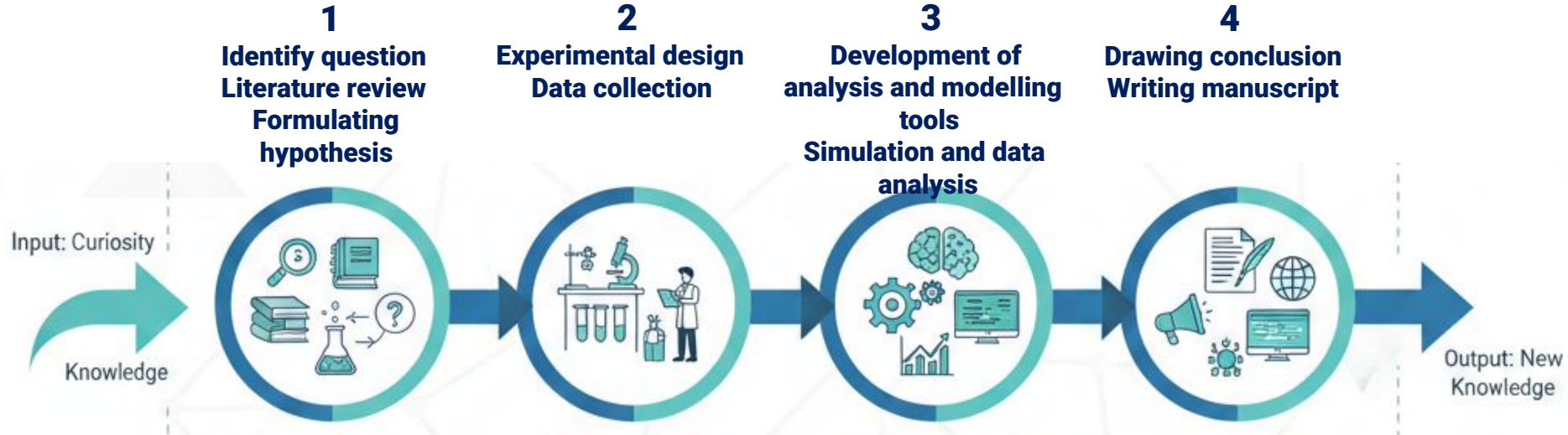
BraiNets
Team



Research Project



Research Project



ChatGPT



Gemini



Claude



Mistral AI



deepseek



Gemini



ChatGPT



Claude

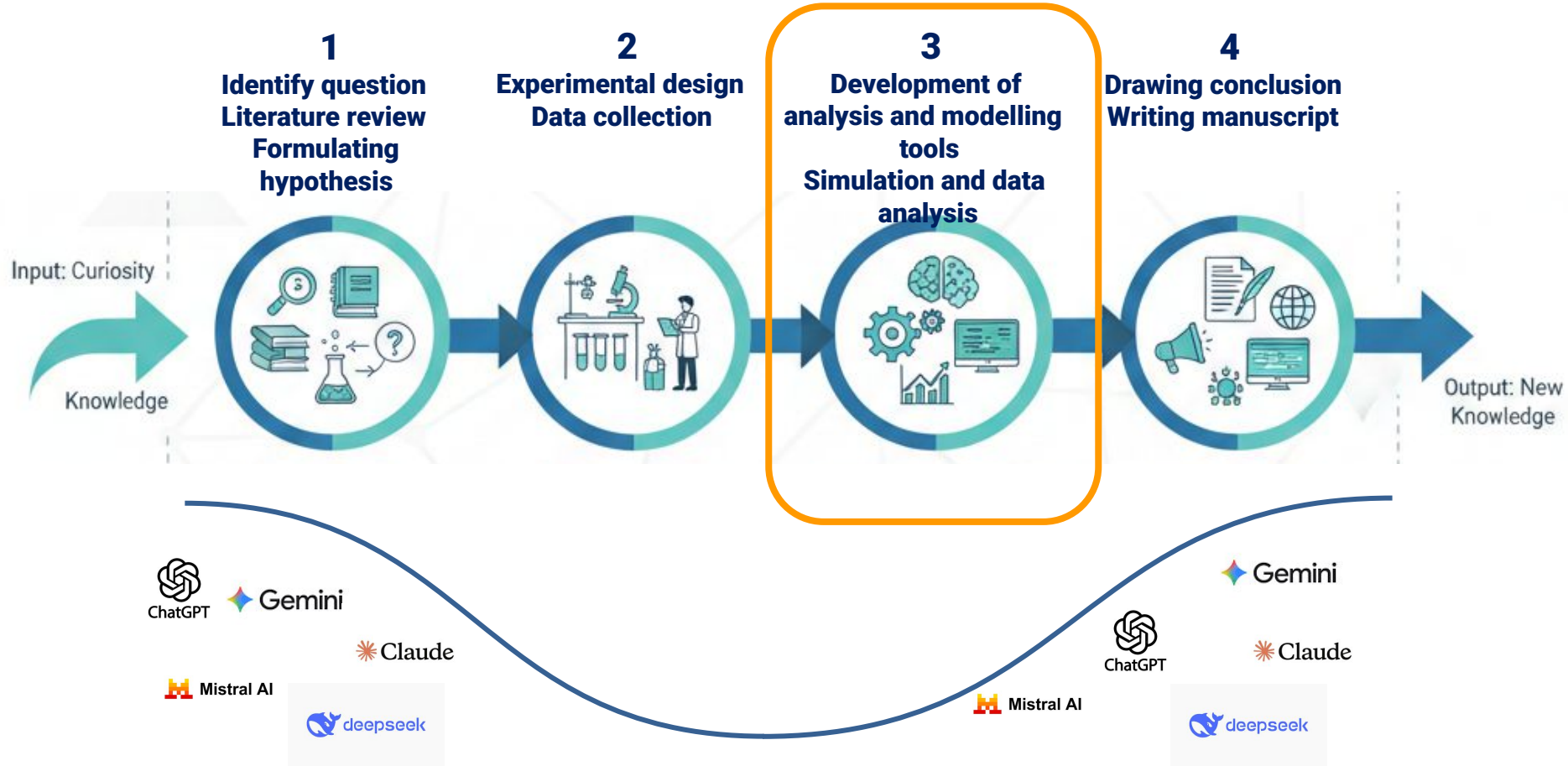


Mistral AI



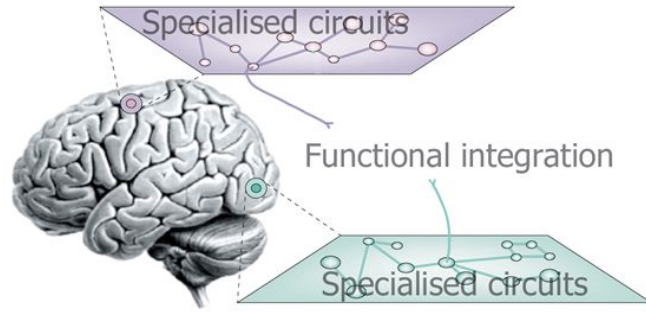
deepseek

Research Project



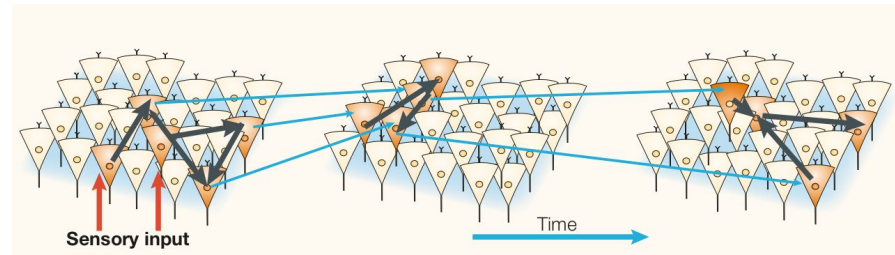
Multiscale Brain Interactions Analysis

The brainweb hypothesis



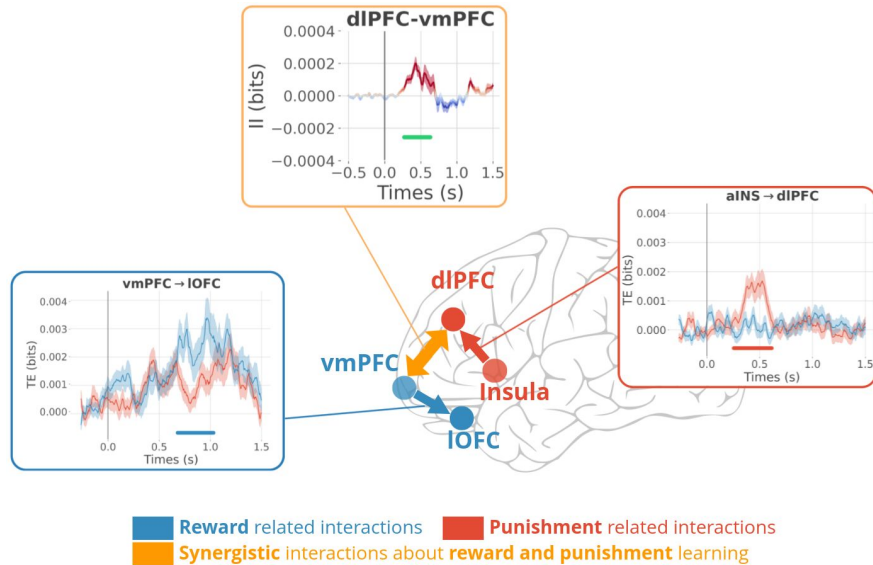
Varela et al (2001) *Nature Reviews Neurosci*

Cell assembly hypothesis (Hebb 1949)



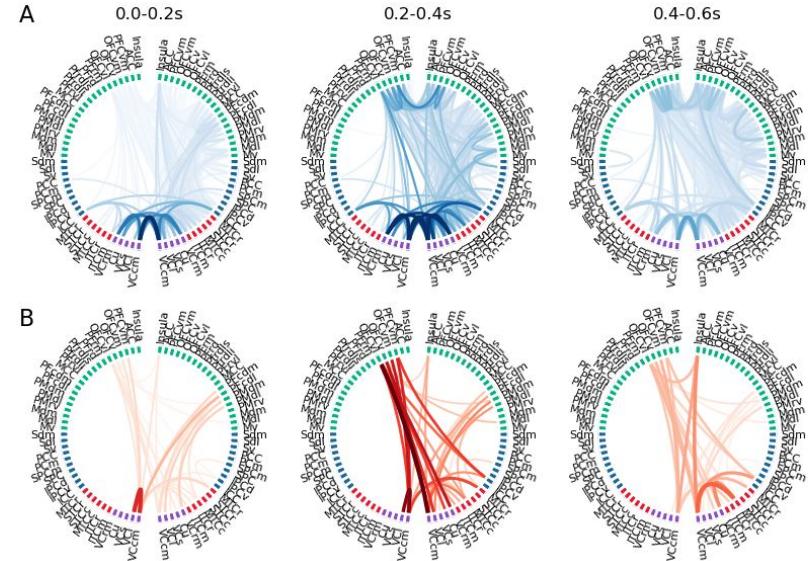
Harris (2005) *Nature Reviews Neurosci*

Neural interactions in the human frontal cortex dissociate reward and punishment learning



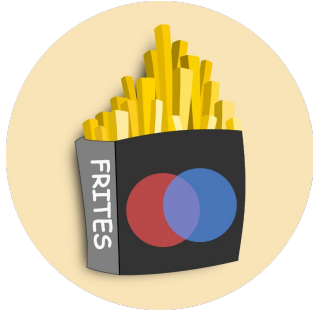
Combrisson et al (2024) *eLife*

Higher-order and synergistic functional interactions encode information gain in goal-directed learning

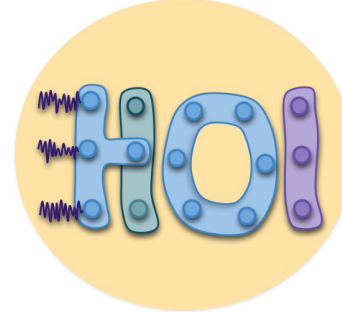


Combrisson et al (2025) *Nat Comms*

BraiNets Python Packages



FRamework for
Information
Theoretical analysis of
Electrophysiological data and
Statistics



Higher
Orders
Interactions



<https://github.com/brainets/frites>



GitHub

<https://brainets.github.io/frites/>

Combrisson et al (2022) *JOSS*



<https://github.com/brainets/hoi>



GitHub

<https://brainets.github.io/hoi/>

Neri et al (2024) *JOSS*



EBRAINS

Dear SantAI,

I hope you are doing well.

Here is my wish list for 2026:

1. Dataset that cannot be shared online (private or large)
→ Local computing
2. Access the latest LLM (GPT 5.2, Gemini 3, etc)
→ Large LM, no small LM
3. Develop, Execute and Correct code → AI Agents, no only chatbot
4. Add domain-specific expertise → Add expertise, no fine-tuning
5. "Free"

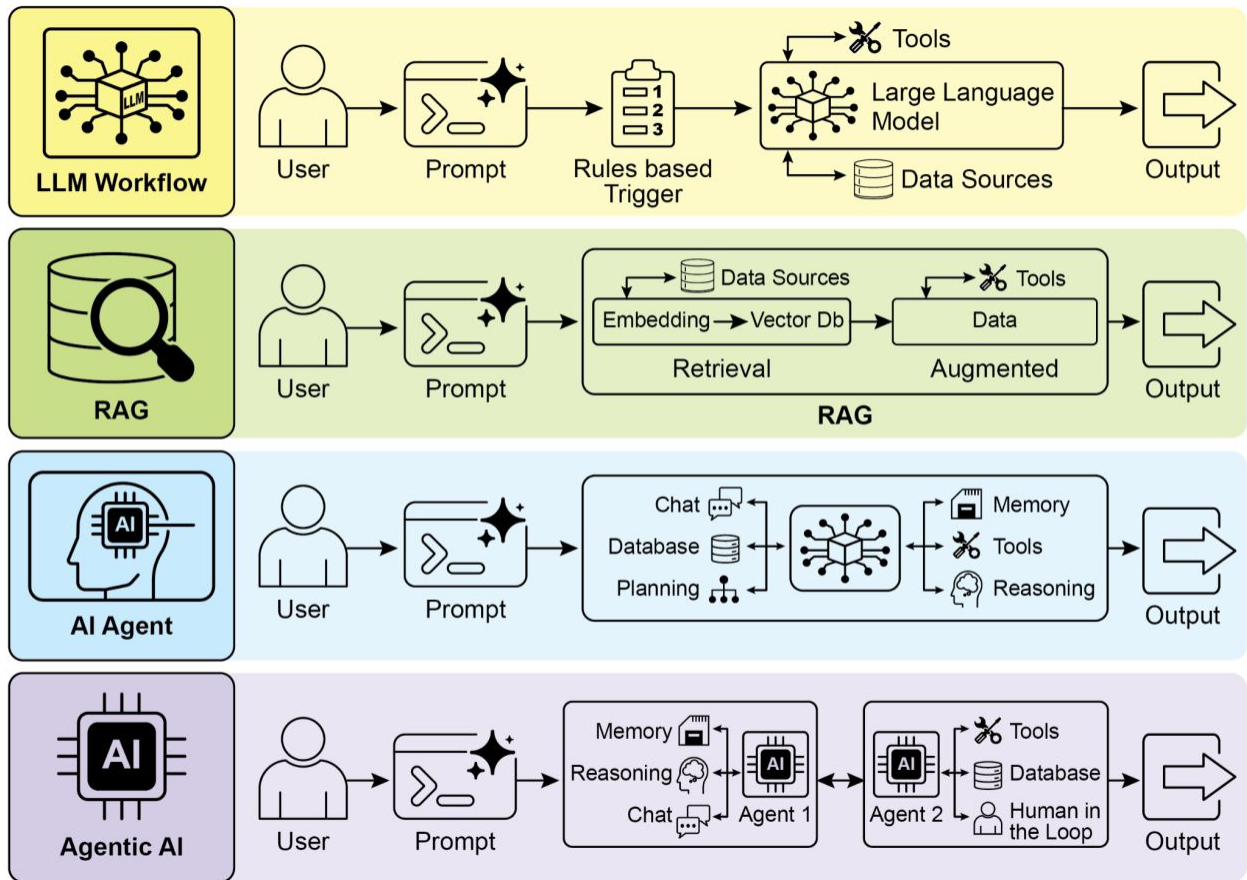
Love from Marseille

Antonio Gullí

Agentic Design Patterns

A Hands-On Guide to Building Intelligent Systems

 Springer



AI agents using Command Line Interface (CLI)

Tool	Coding	Refactoring	Reasoning	Scientific / Ideation	Braina	Free option	Typical paid plans	Positioning
Claude Code CLI	★★★★★	★★★★★	★★★★★	★★★★★	✓	Yes (limited)	Pro ≈ 20/month; Max ≈ 100–200/month	Deep, analytical work
Gemini CLI	★★★★★	★★★★★	★★★★	★★★★★	✓	Yes	Pro ≈ 20/month; Ultra ≈ 250/month	Fast, multimodal, long context
Codex CLI	★★★★★½	★★★★★½	★★★★	★★★★	✓	No	Plus ≈ 20/month; Pro ≈ 200/month	Classical coding assistant
Mistral / Aider / Kimi	★★★★★	★★★★★	★★★★	★★★½	✗	Yes	Optional self-hosting costs	Open, privacy-focused

AI Agent for **Brain Interactions** Analysis



<https://github.com/brainets/braina>

What's inside



braina

Public



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Watch

0



main ▾



1 Branch



0 Tags



Go to file



Add file ▾



<> Code ▾



brovelli update

6f24f12 · last week



36 Commits



docs

updated logo again

last month



examples

added examples from frites and hoi

2 weeks ago



mcp

update

last week



papers

updated README

last month



tutorials

Updated links of frites and hoi

last month



usecases

Added solution of geminicli to master_td

2 weeks ago



GEMINI.md

Added all tutorials, data and papers

last month



LICENSE

Initial commit

last month

Does it work?

Questions of a TD (3h) at the Master in Computational Neuroscience at AMU (2024)

Goal: Verify the agent's ability to simulate autoregressive data and recover directed connectivity patterns using Frites. Prompt:

1. Inspect and take inspiration from the Jupyter notebooks in @braina/examples
2. Generate code and write a standard Markdown (.md) file with clear code blocks of the following problems 3 and 4.
3. Auto-regressive model Simulate a simple autoregressive model composed of a source signal X and a target signal Y (pairwise AR model).
 - a. The signals must oscillated at 40Hz
 - b. Generate 100 trials with a single stimulus and plot the single trial data and causal coupling over time. Plot the time series using the plot() function
 - c. Generate 100 trials with a three stimuli and plot the single trial data and causal coupling over time
 - d. Write the equation of the AR model generating the data. Hint: look at the source code
4. Single-trial time-resolved (dynamic) Functional Connectivity Take the data simulated in 1b and estimate dynamic functional connectivity between X and Y with the following parameters: window length = 0.75 seconds, win_step = 0.02 seconds
 - a. Computed the DFC for each trial and plot the average time course across trials
 - b. Test different window_length values and plot different examples, try to interpret the results
5. Convert the md file using jupyterx into a Jupyter notebook

Solution (Brovelli)

← Files

main

braina / usecases / master_td / solutions.ipynb

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PreviewCodeBlame

2) Single-trial time-resolved (dynamic) Functional Connectivity

Now, we take the data simulated in 1b and estimate dynamic functional connectivity between X and Y with the following parameters: window length = 0.75 seconds, win_step = 0.02 seconds

a) Computed the DFC for each trial and plot the average time course across trials

b) Test different window_length values and plot different examples, try to interpret the results

In [144]:

```
# Define windows
slwin_len = .7 # 500ms window length
slwin_step = .02 # 20ms between consecutive windows
win_sample = define_windows(ar.times.data, slwin_len=slwin_len,
                             slwin_step=slwin_step)[0]
```

Defining temporal windows (sfreq=199.99999999999983)

Definition of sliding windows (len=0.7, start=-0.5, stop=1.495, step=0.02)

In [145]:

```
# compute the DFC on sliding windows
dfc = conn_dfc(ar, times='times', roi='roi', win_sample=win_sample)
# compute DFC
#dfc = conn_dfc(ar, win_sample, times=ar.times.data, roi=ar.roi)

plt.figure(figsize=(7, 5))
# plt.plot(times p, dfc.mean('trials').T)
dfc.mean('trials').plot(x='times', hue='roi')
plt.xlabel('Time')
plt.title("Mean DFC across trials between pairs of roi")
plt.grid()
plt.show()
```

Defining links (n_roi=2; directed=False; net=False; nb_min_links=None)

Sorting roi names

Gaussian Copula Mutual Information Estimator (mi_type=cc, copnorm=False, biascorrect=False, demeaned=False)

github.com x

Solution (Braina)

← Files

main

braina / usecases / master_td / geminici_solutions.ipynb

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4. Single-trial time-resolved (dynamic) Functional Connectivity

We estimate the Dynamic Functional Connectivity (DFC) between X and Y using the data from the single-stimulus simulation (3b).

4a. Compute DFC

We use a sliding window approach with window_length = 0.75s and win_step = 0.02s.

In [4]:

```
# Define parameters
slwin_len = 0.75 # Window length in seconds
slwin_step = 0.02 # Step size in seconds

# Define sliding windows
win_sample = define_windows(times, slwin_len=slwin_len, slwin_step=slwin_step)[0]

# Compute DFC
dfc = conn_dfc(ar, win_sample, times=times, roi=ar.roi.data, n_jobs=1)

# Plot average DFC across trials
dfc_mean = dfc.mean('trials')

plt.figure(figsize=(10, 6))
for roi_pair in dfc_mean.roi.data:
    plt.plot(dfc_mean.times, dfc_mean.sel(roi=roi_pair), label=roi_pair)

plt.title("Average Dynamic Functional Connectivity (X-Y)")
plt.xlabel("Time (s)")
plt.ylabel("Mutual Information (MI)")
plt.legend()
plt.show()
```

Defining temporal windows (sfreq=199.99999999999983)



AI

**CODE
MACHINE
3000**

**TECH
DEBT**

**HOW TO
FIX
AI-GENERATED
CODE**



AI Agent for **Brain Interactions** Analysis



<https://github.com/brainets/braina>